AI-Based Deepfake Detection System

Problem Statement

The rapid advancement of Artificial Intelligence and Deep Learning technologies has increased the creation and spread of deepfake content across digital platforms. Deepfakes are manipulated or AI-generated multimedia contents such as fake images, videos, audio recordings, and misleading text that appear realistic and authentic.

These fake contents are widely used for misinformation, cybercrime, identity theft, financial fraud, fake news generation, and social manipulation. Existing deepfake detection systems mainly focus on a single media type and fail to provide accurate detection for multiple multimedia formats together.

The proposed AI-Based Deepfake Detection System supports multiple input formats including text, image, audio, and video. If the input is text, the system analyzes the content using Natural Language Processing (NLP) techniques to identify misleading, manipulated, or AI-generated text. If the input is an image or video, the system uses Computer Vision and Deep Learning techniques to detect visual manipulation and deepfake patterns. For audio inputs, speech analysis and voice pattern recognition are used to identify fake or cloned voices.

The system automatically processes the given input type, analyzes its authenticity, and provides accurate deepfake detection results. The project aims to improve digital security, media authenticity, and public trust using AI-based automated multimedia analysis.

Objectives

- To develop an AI-based system for detecting deepfake content.
- To support multiple input formats such as text, image, audio, and video.
- To analyze text using Natural Language Processing techniques.
- To detect manipulated images and videos using Computer Vision and Deep Learning.
- To identify fake or cloned voices using speech analysis techniques.
- To improve digital security and prevent misinformation spread.
- To provide accurate and reliable deepfake detection results.